

Introduction to Basic Statistics

With real world healthcare examples (MIMIC-III)



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Why does statistics matter in healthcare data?

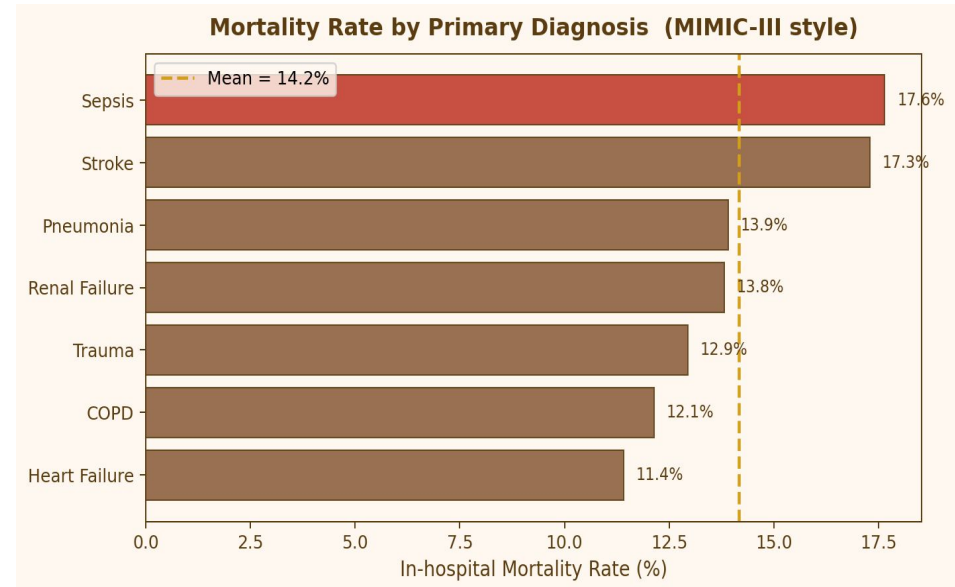


12% of ICU patients in MIMIC-III did not survive.
Without statistics, we cannot identify WHY.

Same data. Wrong analysis.
Different (and dangerous) conclusions.

Usage of Statistics

- ▶ **Signal processing** → noise analysis
- ▶ **Quality control** → process variation
- ▶ **Machine learning** → model evaluation
- ▶ **Clinical trials** → drug efficacy
- ▶ **A/B testing** → product decisions



Types of data



- Knowing the data before analyzing

QUANTITATIVE (Numerical)

► Discrete — countable, whole numbers

- # Admissions per day: 3, 7, 12
- # Blood transfusion units

► Continuous — any value in a range

- Heart rate: 72.4 bpm
- Temperature: 37.2 °C
- Length of Stay: 4.7 days

QUALITATIVE (Categorical)

► Nominal — no natural order

- Blood type: A, B, AB, O
- Gender: Male / Female
- ICU type: MICU, SICU, CCU

► Ordinal — ordered categories

- Pain scale: Mild → Moderate → Severe
- SOFA score: 0–24

(SOFA - Sequential Organ Failure Assessment)

Python statistics tool-kit



NumPy

Arrays, math ops,
broadcasting

Pandas

DataFrames, groupby, merge

Matplotlib

Base plotting library

Seaborn

Statistical visualisations

SciPy

Stats tests, distributions

Statsmodels

Regression, ANOVA, time
series

Install everything: `pip install numpy pandas matplotlib seaborn scipy statsmodels`

Measure of central tendency



- Where is the center of your data?

Mean ($\bar{x}$)

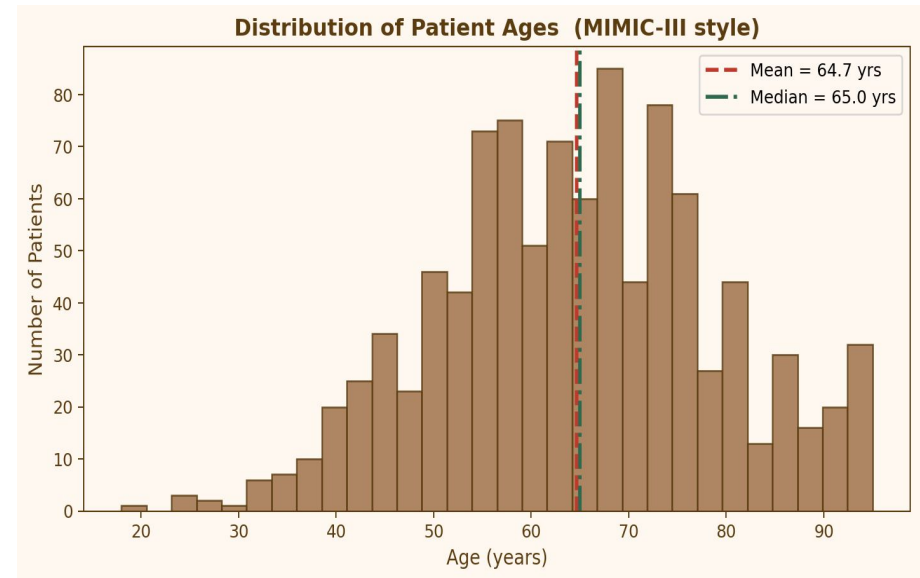
Sum of all values $\div$ n

Median

Middle value after sorting

Mode

Most frequently occurring value



$$\bar{x} = (x_1 + x_2 + \dots + x_n) / n$$

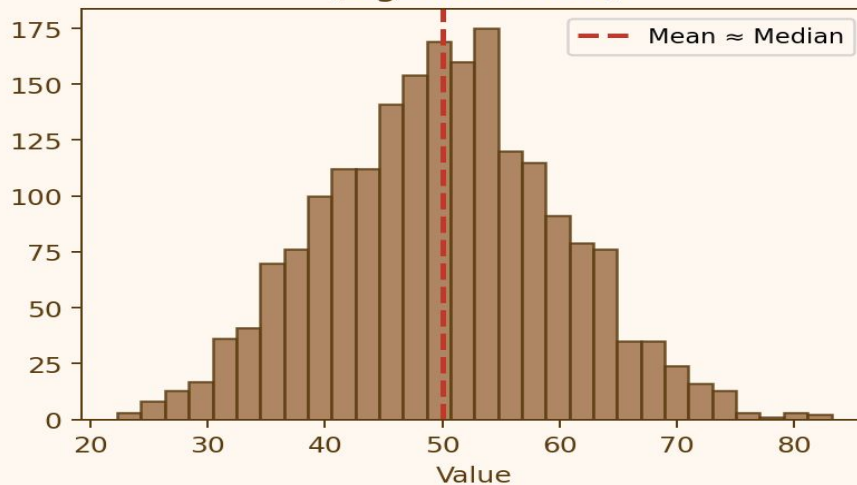
Mean vs Median – which should we use?



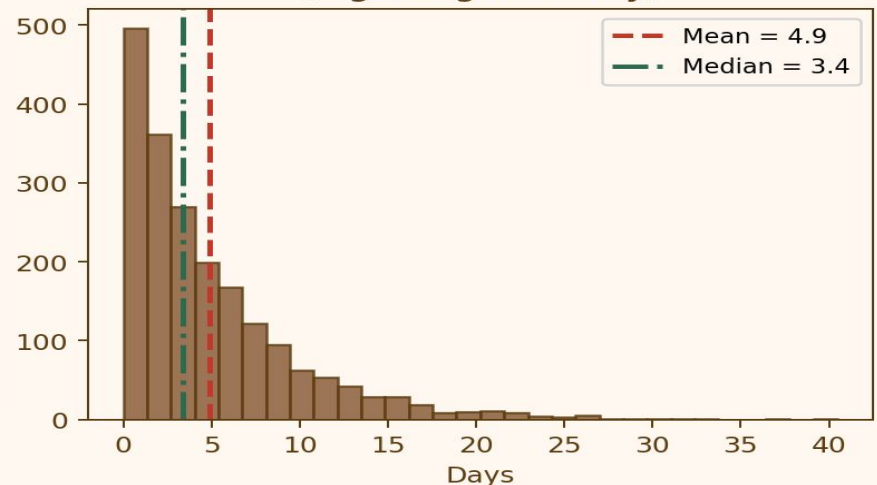
- Answer depends on your data shape

Mean vs Median — Why it Matters

**Symmetric Distribution
(e.g. Heart Rate)**



**Right-Skewed Distribution
(e.g. Length of Stay)**



Symmetric → Mean $\approx$ Median
Use Mean (e.g. heart rate, BP)

Skewed → Mean $\neq$ Median
Use Median (e.g. LOS, salary data)

**A billionaire walks into a bar. Everyone's average salary jumps to ₹10 Crore.
But the median barely moves. → Median is robust to outliers.**

Measure of spread (Dispersion)



- The mean tells half of the story – spread tells the rest

► **Range = Max – Min**

- Quick but sensitive to outliers

► **Variance (σ^2) = $\Sigma(x - \bar{x})^2 / n$**

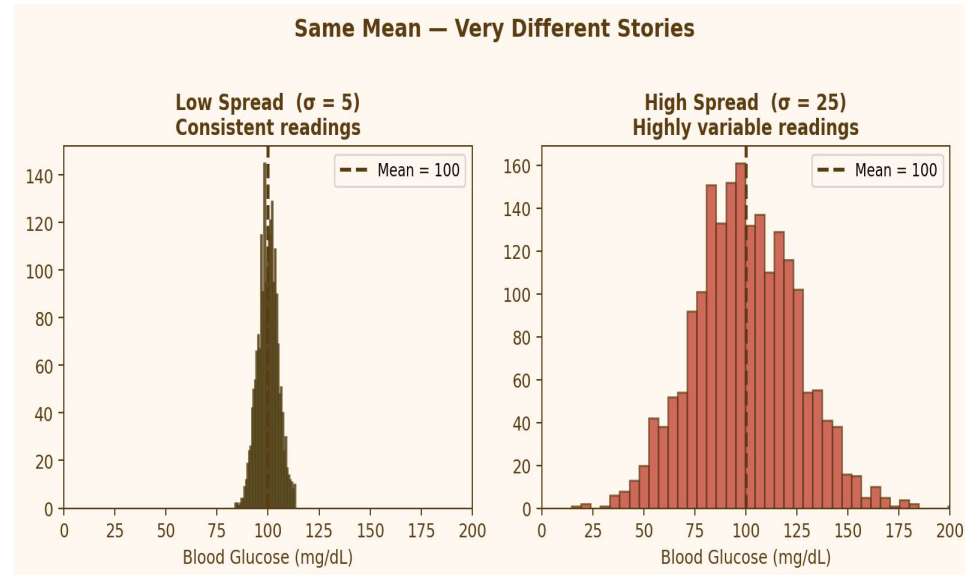
- Average squared distance from mean

► **Std Deviation (σ) = $\sqrt{\text{Variance}}$**

- Same unit as data → most interpretable

► **IQR = Q3 – Q1**

- Robust spread: covers middle 50% of data



MIMIC-III: Blood Glucose Stats

- Mean: ~120 mg/dL
- Std Dev: ~40 mg/dL
- Range: 50–400 mg/dL
- IQR: ~55 mg/dL (Q1~90, Q3~145)

Analogy — Archery Target

Archery Analogy:

- Low σ = arrows clustered tightly (consistent)
- High σ = arrows scattered widely (variable)
- Two ICUs can have the same avg BP but very different σ !

The five number summary & box plot



- The data's report card in single glance

Minimum

Lowest non-outlier value

Q1 (25th Pctl)

Lower quartile

Median (Q2)

Middle value — 50% below, 50% above

Q3 (75th Pctl)

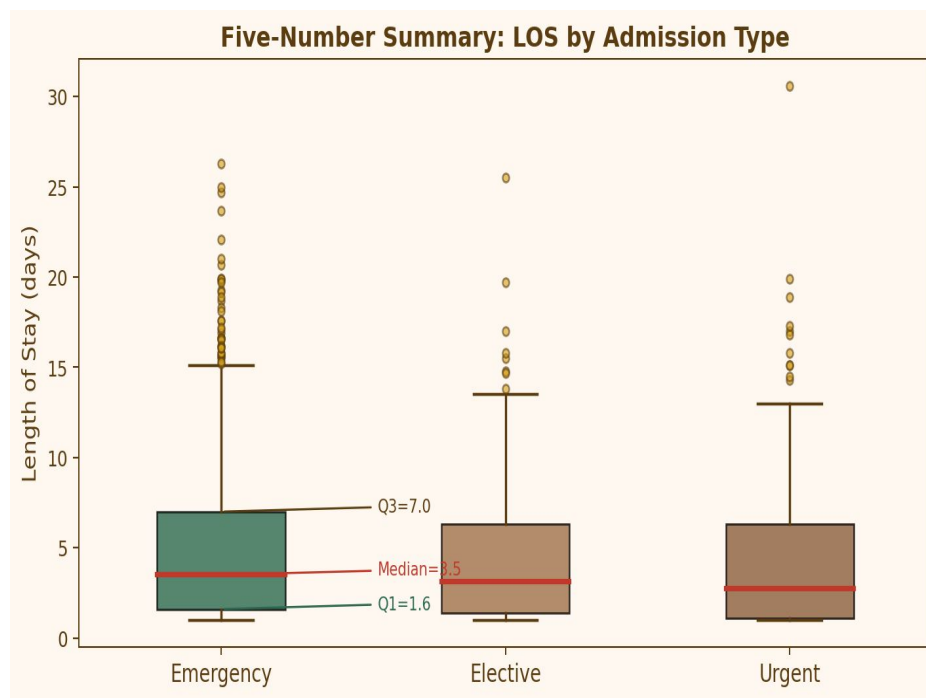
Upper quartile

Maximum

Highest non-outlier value

Outliers ♦

Points beyond $1.5 \times \text{IQR}$ from Q1/Q3



Why Box Plots?

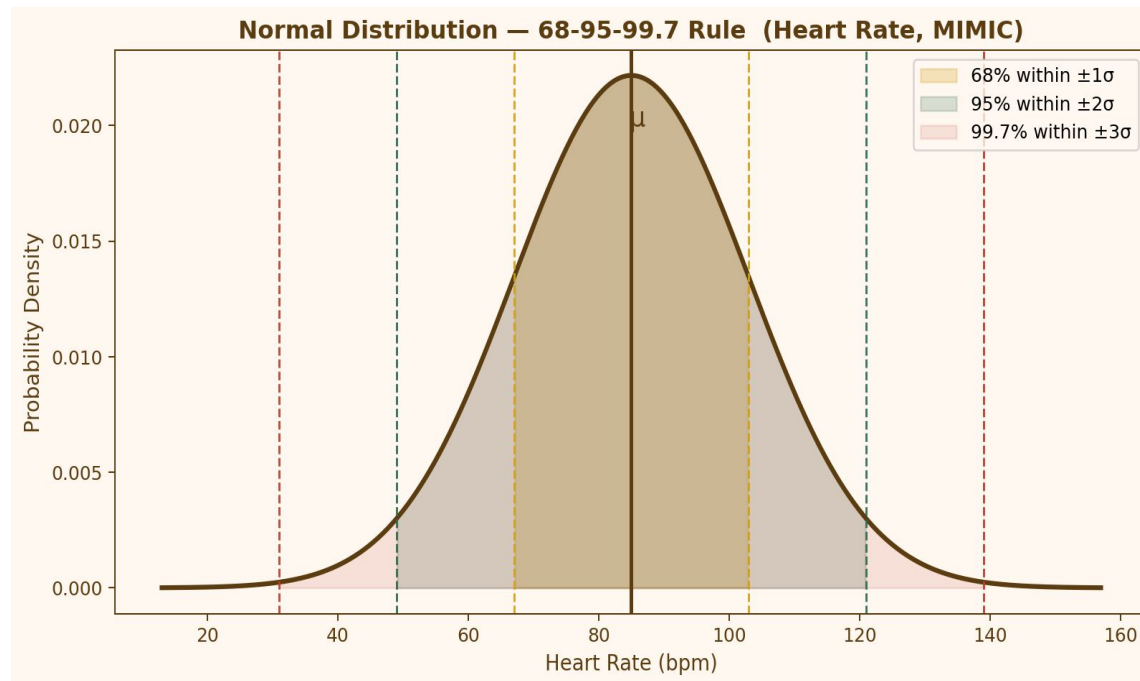
Box plots are ideal for comparing distributions across groups.
Emergency vs Elective vs Urgent admissions — who stays longer?

Normal distribution



- ▶ 68% of patients have HR within $\pm 1\sigma$
- ▶ 95% of patients have HR within $\pm 2\sigma$
- ▶ 99.7% of patients have HR within $\pm 3\sigma$
- ▶ Normal (Gaussian) distribution occurs when:
 - Many small, independent factors add up
 - Examples: Heights, IQ scores, HR, BP
- ▶ In MIMIC-III: Heart Rate $\sim N(85, 18^2)$ bpm

Normality check: Shapiro-Wilk test | Q-Q plot | Histogram visual inspection



Correlation



- Measuring the strength and direction of a linear relationship

► **Pearson's r ranges from -1 to $+1$**

- $r = +1$: perfect positive relationship
- $r = 0$: no linear relationship
- $r = -1$: perfect negative relationship

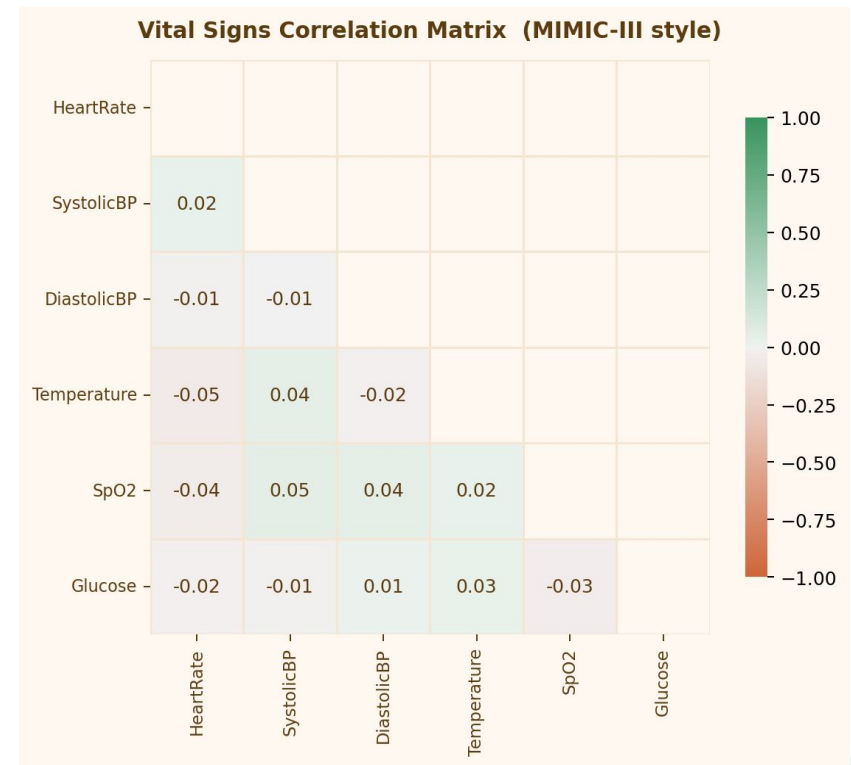
► **Rule of thumb (for $|r|$):**

- $0.0 - 0.3 \rightarrow$ Weak
- $0.3 - 0.7 \rightarrow$ Moderate
- $0.7 - 1.0 \rightarrow$ Strong

Correlation $\neq$ Causation!

Classic Trap: Ice cream sales & drowning rates both peak in summer.

$r \approx +0.9$ — but ice cream does NOT cause



Systolic BP vs Diastolic BP in MIMIC-III: $r \approx +0.65$ (Moderate-Strong)
Age vs Creatinine: $r \approx +0.15$ (Weak positive — kidneys age with us)

Hypothesis testing



- Making evidence-based decision from data

The Courtroom Analogy

Courtroom Analogy:

H_0 (Null) = Defendant is INNOCENT until proven otherwise

H_1 (Alternate) = Defendant IS GUILTY

p-value = strength of evidence against H_0

Verdict: $p < 0.05 \rightarrow$ Reject H_0 (significant evidence)

- ▶ 1. State H_0 and H_1
- ▶ 2. Choose significance level ($\alpha = 0.05$)
- ▶ 3. Collect data and compute test statistic
- ▶ 4. Calculate p-value
- ▶ 5. Compare p-value with $\alpha \rightarrow$ decide

p-value	Decision	Meaning
$p < 0.001$	Reject H_0	Very strong evidence
$p < 0.05$	Reject H_0	Statistically significant
$p \geq 0.05$	Fail to reject H_0	Insufficient evidence

Two Types of Errors

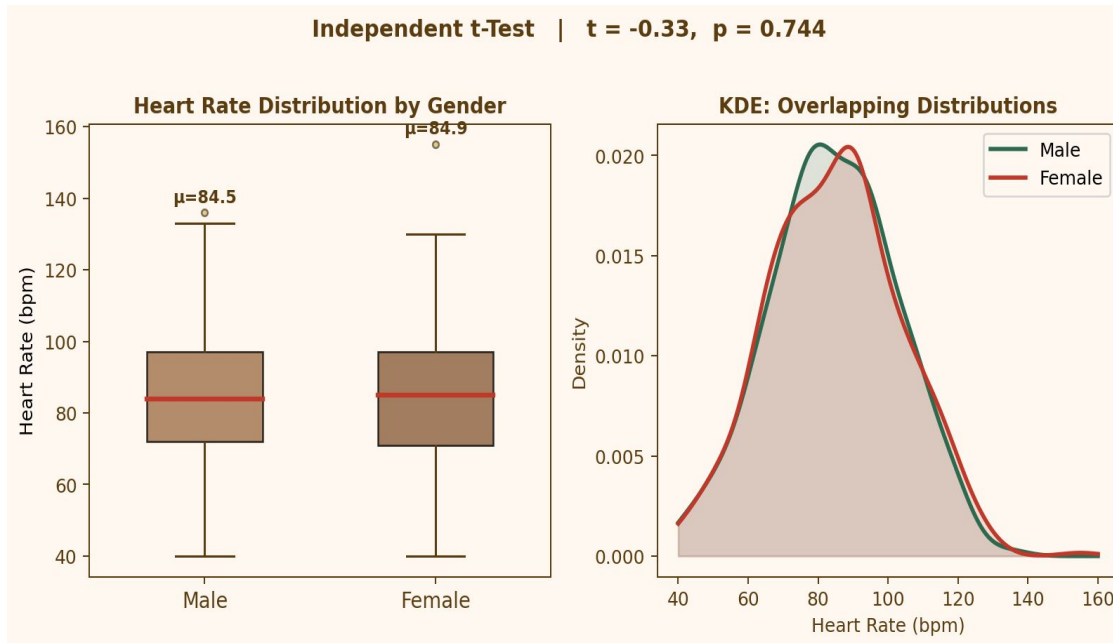
Type I Error (α): Convict an innocent person

Type II Error (β): Acquit a guilty person
We control α (usually 0.05) when designing the test.

T-test in action



- Is the observed difference real or just random chance?



Independent t-Test

- ▶ $H_0: \mu_{\text{Male}} = \mu_{\text{Female}}$
- ▶ $H_1: \mu_{\text{Male}} \neq \mu_{\text{Female}}$
- ▶
- ▶ Male HR: $\mu \approx 85.2$ bpm
- ▶ Female HR: $\mu \approx 84.7$ bpm
- ▶
- ▶ t-statistic: ~ 0.45
- ▶ p-value: > 0.05
- ▶
- ▶ **Conclusion: Fail to reject H_0**
- ▶ **No significant gender difference**

- ▶ Use when comparing means of TWO groups
- ▶ Assumes approx. normal distribution
- ▶ Or large sample ($n > 30$) by CLT
- ▶ t-statistic = $(\bar{x}_1 - \bar{x}_2) / SE$
- ▶ SE = standard error of difference

Chi square test



- Are two categorical variables are associated or independent?

MIMIC-III: Gender × Mortality Contingency Table

	Survived	Died	Total
Male	~487	~63	~550
Female	~393	~57	~450
Total	~880	~120	~1000

► Formula:

$$\chi^2 = \sum (\text{Observed} - \text{Expected})^2 / \text{Expected}$$

► In MIMIC-III example:

- $\chi^2 \approx 0.22$, $df = 1$
- $p \approx 0.64 \rightarrow$ Fail to reject H_0
- Conclusion: Gender is NOT significantly associated with mortality in this sample

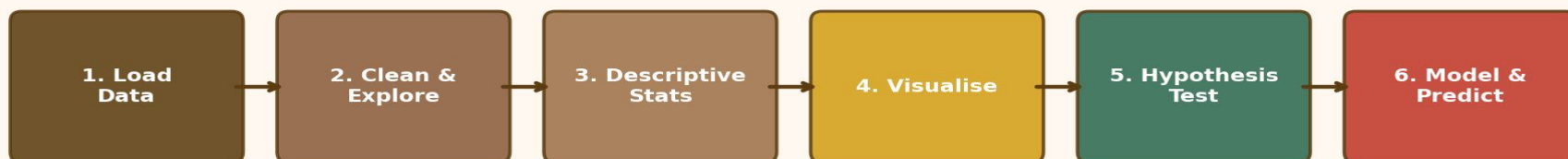
Analogy — Fair Coin

Coin flip analogy: Toss a fair coin 100 times.
You expect 50 H and 50 T. If you get 70 H, is it biased?
Chi-square compares OBSERVED vs EXPECTED frequencies.

Putting it all together



Statistical Analysis Workflow with MIMIC-III



Each step is demonstrated in the accompanying Jupyter Notebook

Key Takeaways



1

Always visualise first.

A histogram or box plot reveals what numbers alone cannot.

2

Mean $\neq$ always the right average.

For skewed data (salary, LOS), prefer the median.

3

Spread matters as much as centre.

Two datasets with the same mean can behave very differently.

4

$p < 0.05$ is not magic.

Statistical significance $\neq$ practical significance. Report effect size too.

5

Correlation is NOT causation.

Always ask: is there a confounding variable explaining both trends?

THANK YOU



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